

Planetary Defense: AI-driven Asteroid Classification

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Abstract

Near-Earth Objects (NEOs) represent a non-negligible risk for planetary safety, making their detection and classification a central task in modern space monitoring. This study proposes a machine learning framework for identifying potentially hazardous asteroids using observational data from the NASA NEO dataset. The dataset contains more than 90,000 observations describing the physical and orbital characteristics of asteroids. The analytical pipeline includes data preprocessing, feature standardization, and the development of two classification models: a baseline Logistic Regression and an Artificial Neural Network (ANN). Particular attention is given to the strong class imbalance present in the dataset. By integrating class weighting strategies and focusing on recall optimization for hazardous objects, the proposed approach aims to minimize false negatives and improve automated planetary defense monitoring.

1 Introduction

Monitoring **Near-Earth Objects** has become an essential component of planetary defense. Although most asteroids pass safely near Earth, historical events demonstrate that asteroid impacts can cause significant damage. Events such as the Tunguska explosion in 1908 and the Chelyabinsk meteor in 2013 illustrate how even relatively small objects can generate large-scale consequences when entering the Earth's atmosphere. Modern astronomical surveys continuously detect thousands of new objects and generate massive amounts of observational data. While these systems are highly effective at discovering asteroids, the main challenge lies in distinguishing harmless objects from those that may pose a potential threat. As the number of detected NEOs increases, automated analytical systems based on Artificial Intelligence become increasingly important to support large-scale monitoring. This work investigates how machine learning models can assist in the classification of hazardous asteroids using physical and orbital parameters derived from observational data. The research addresses the following questions:

- **RQ1:** *Can machine learning models effectively dis-*

tinguish hazardous from non-hazardous asteroids using basic orbital and physical features?

- **RQ2:** *How does class imbalance affect model performance in asteroid hazard classification?*
- **RQ3:** *Does increasing the depth of a Neural Network significantly improve the separation capability for hazardous objects compared to linear baseline models?*

The main **contributions** of this study are:

- Development of an **analytical pipeline** for asteroid hazard classification based on the NASA NEO dataset [1].
- Evaluation of Gradient Descent for **Logistic Regression** as a baseline model.
- Implementation of **Artificial Neural Networks** capable of capturing nonlinear relationships between asteroid characteristics.
- Analysis of **class imbalance** effects and the application of cost-sensitive learning to improve detection of hazardous objects.

2 Related Work

Several recent studies have explored the application of machine learning techniques to asteroid hazard classification.

A study published in ScienceDirect evaluates multiple decision trees algorithms for asteroid classification, including Random Forest, Extra Trees, LightGBM, Gradient Boosting, and AdaBoost [2]. Using observational asteroid data, the authors compare models using ROC-AUC metrics and report that Random Forest achieved the highest performance with an AUC of approximately 0.94. This work highlights the effectiveness of ensemble tree methods for extracting complex patterns from astronomical datasets.

Another recent approach proposes a deep learning framework based on Graph Neural Networks (GNNs) for asteroid hazard classification [3]. In this system, asteroids are represented as nodes within a graph and

relationships between objects are modeled through feature similarity. The architecture allows the model to capture structural relationships among asteroids, achieving an overall accuracy close to 99% and an AUC near 0.99. However, despite the high global accuracy, the recall for hazardous asteroids remains significantly lower (around 78%), even when oversampling strategies such as SMOTE are applied. While these approaches achieve strong global classification metrics, they primarily focus on maximizing overall accuracy or AUC.

In contrast, the present work prioritizes maximizing recall for the hazardous class, since failing to detect a dangerous object represents the most critical error in planetary defense systems. The proposed framework therefore focuses on cost-sensitive learning strategies and evaluates how class imbalance influences model behavior.

3 Methodology and System Design

The proposed framework follows a structured analytical pipeline designed to transform raw observational data into a reliable binary classification system for asteroid hazard detection.

3.1 Analytical Pipeline

The pipeline consists of the following sequential stages:

- **Data Cleaning:** Non-informative attributes (e.g., identifiers and nominal labels) are removed, along with zero-variance features that do not contribute to discrimination.
- **Target Encoding:** The target variable is converted into a binary format: $y = 1$ Hazardous, 0 Safe.
- **Feature Standardization:** All numerical features are normalized using Z-score standardization to ensure homogeneous scaling:

$$x_{std} = \frac{x - \mu}{\sigma}. \quad (1)$$

- **Dataset Splitting:** The dataset is divided using a stratified splitting strategy to preserve class proportions across subsets. The data is partitioned into training, validation, and test sets, ensuring that the minority (hazardous) class is adequately represented in each subset. This is essential to obtain reliable performance estimates and avoid biased evaluation.
- **Model Definition Strategy:** A Logistic Regression baseline is first established to provide a performance benchmark. Subsequently, a neural model is designed using a sequential architecture to capture more complex relationships through progressive transformation of input features.

- **Handling Class Imbalance:** Due to the strong imbalance between hazardous and non-hazardous asteroids, a cost-sensitive learning approach is adopted. This is implemented by modifying the standard Binary Cross-Entropy loss function through **class weighting**:

$$L = -w_1 \cdot y \log(p) + w_0 \cdot (1 - y) \log(1 - p), \quad (2)$$

where p is the predicted probability, y is the actual label, w_1 is the weight for the hazardous class, and w_0 is the weight for the safe class. The class weights are computed inversely proportional to class frequencies:

$$w_c = \frac{N}{2 \cdot N_c}, \quad (3)$$

where N is the total number of samples, and N_c is the number of samples in class c . This formulation assigns a higher penalty to errors on the minority class, forcing the model to focus more on correctly identifying hazardous asteroids and reducing false negatives.

3.2 System Design

3.2.1 Logistic Regression (Baseline)

Logistic Regression estimates the probability of an asteroid being hazardous through the sigmoid function:

$$\text{sig}(z) = \frac{1}{1 + e^{-z}}, \quad (4)$$

where

$$z = w^T x + b. \quad (5)$$

Two configurations are evaluated. The first is a standard Logistic Regression model without any correction for class imbalance. The second introduces class weighting, penalizing errors on the minority class.

3.2.2 Artificial Neural Network

The architecture consists of an input layer receiving standardized features, followed by one or two fully connected hidden layers with nonlinear activation functions, and a final output layer producing a probability score. The model is evaluated under two different configurations. The first does not include any imbalance correction, allowing observation of the network’s natural learning behavior. The second incorporates class weighting, forcing the model to place greater emphasis on correctly identifying hazardous asteroids. Additionally, different architectural configurations are explored to assess the impact of model depth and capacity on classification performance.

3.3 Evaluation Metrics

Given the imbalanced nature of the dataset, model performance is evaluated using metrics that go beyond simple accuracy. Let: TP be **True Positives** (hazardous correctly identified), TN be **True Negatives** (safe correctly identified), FP be **False Positives** (safe classified as hazardous), and FN be **False Negatives** (hazardous classified as safe).

The following metrics are computed:

- **Accuracy** — overall proportion of correctly classified samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}; \quad (6)$$

- **Recall** (Sensitivity) – Hazardous Class — ability to correctly detect hazardous asteroids:

$$\text{Recall} = \frac{TP}{TP + FN}; \quad (7)$$

- **Precision** – Hazardous Class — reliability of hazardous predictions:

$$\text{Precision} = \frac{TP}{TP + FP}; \quad (8)$$

- **F1-Score** — harmonic mean of precision and recall, balancing detection and reliability:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}; \quad (9)$$

- **Matthews Correlation Coefficient** (MCC) — balanced measure accounting for all confusion matrix terms:

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}; \quad (10)$$

- **Cohen’s Kappa** — agreement between predictions and true labels corrected for chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad (11)$$

where p_o is the observed agreement

$$p_o = \frac{TP + TN}{TP + TN + FP + FN}, \quad (12)$$

and p_e is the expected agreement by chance

$$p_e = \frac{(TP + FP)(TP + FN) + (TN + FP)(TN + FN)}{(TP + TN + FP + FN)^2}. \quad (13)$$

Among these metrics, recall for the hazardous class is considered the most critical, as it quantifies the ability of the system to correctly identify potentially dangerous asteroids and minimize false negatives.

4 Experimental Setup and Results

4.1 Experimental Setup

4.1.1 Dataset Description

The experiments are conducted on the NASA Near-Earth Object dataset, which includes 90,836 asteroid observations collected through astronomical surveys. Each observation is described by a set of physical and orbital features, including absolute magnitude, estimated diameter, relative velocity, and miss distance (Tab1). A key characteristic of the dataset is the strong class imbalance: approximately 90.3% of the asteroids are classified as Safe, while only 9.7% are labeled as Hazardous. To ensure a reliable and unbiased evaluation, the dataset is partitioned using a stratified splitting strategy, preserving class proportions across all subsets. The data is divided into 70% for training, 15% for validation, and 15% for testing.

4.1.2 Baseline Model: Logistic Regression

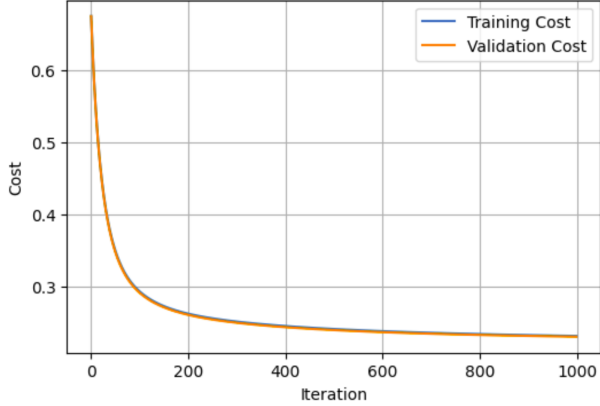
Initially, a custom implementation of the model was developed using a Gradient Descent optimization algorithm. The sigmoid activation function, the binary cross-entropy cost function, and the gradient calculations were

Table 1: Sample of the NASA Near-Earth Object dataset.

id	name	absolute_magnitude	est_diameter_min	est_diameter_max	relative_velocity	miss_distance	orbiting_body	sentry_object	hazardous
2162635	162635 (2000 SS164)	16.73	1.198271	2.679415	13569.24922	54839744.08	Earth	False	False
2277475	277475 (2005 WK4)	20.00	0.265800	0.594347	73588.72666	61438126.52	Earth	False	True
2512244	512244 (2015 YE18)	17.83	0.722030	1.614507	114258.69210	49798724.94	Earth	False	False
3596030	(2012 BV13)	22.20	0.096506	0.215794	24764.30314	25434972.72	Earth	False	False
3667127	(2014 GE35)	20.09	0.255009	0.570217	42737.73376	46275567.00	Earth	False	True
54138696	(2021 GY23)	24.32	0.036354	0.081291	34297.58778	40585691.23	Earth	False	False
54189957	(2021 PY40)	20.95	0.171615	0.383743	27529.47231	29069121.42	Earth	False	False
54230078	(2021 XD6)	28.49	0.005328	0.011914	57544.47008	55115019.26	Earth	False	False

explicitly defined. The training process was performed for 1000 iterations with a learning rate of 0.1, monitoring both the training and validation loss in order to observe the convergence behavior of the optimization process (Fig1).

Figure 1: Training and validation cost vs. iteration for the custom Logistic Regression model.



However, due to the strong class imbalance in the dataset, the initial model showed a clear tendency to predict the majority class (“Safe”). Although this configuration produced a relatively high overall accuracy, it resulted in a very low recall for the “Hazardous” class, which represents the priority target from an application perspective (Tab2).

Table 2: Classification report for the custom Logistic Regression model.

	precision	recall	f1-score	support
0	0.90	1.00	0.95	12299
1	0.37	0.01	0.03	1326
accuracy			0.90	13625
macro avg	0.64	0.51	0.49	13625
weighted avg	0.85	0.90	0.86	13625

To address this limitation, the Logistic Regression model was subsequently implemented using the **Scikit-learn** library with the parameter `class_weight = "balanced"`. The resulting model demonstrated a substantial improvement in detecting potentially hazardous asteroids (Fig2), achieving a recall of approximately 0.93 for the “Hazardous” class, although with a moderate reduction in overall accuracy. The final evaluation on the test set produced an overall accuracy of approximately 78%, with a Matthews Correlation Coefficient of about 0.45 and a Cohen’s Kappa score of approximately 0.36 (Tab3). These values indicate a moderate but meaningful predictive capability, clearly outperforming naive baselines or models that predominantly predict the majority class.

4.1.3 Advanced Model: Artificial Neural Network

The Neural Network was developed using the TensorFlow/Keras framework and trained on the standardized

Figure 2: Confusion matrix of the Logistic Regression model with class weighting.

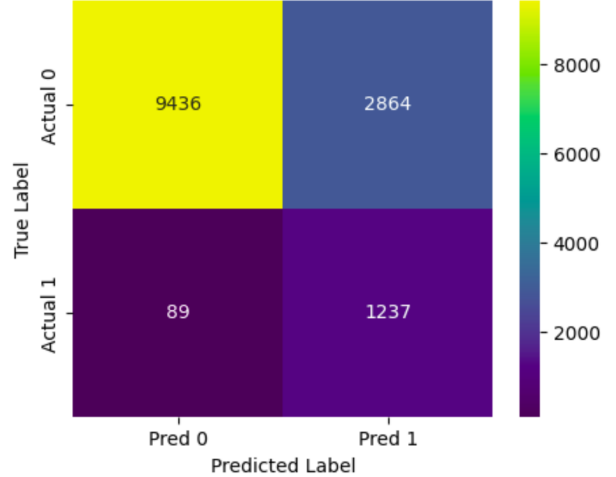


Table 3: Performance metrics and Classification report for the Logistic Regression model using balanced class weights.

Matthews Correlation Coefficient (MCC): 0.452
Cohen's Kappa: 0.362

	precision	recall	f1-score	support
Safe (0)	0.99	0.77	0.86	12300
Hazardous (1)	0.30	0.93	0.46	1326
accuracy			0.78	13626
macro avg	0.65	0.85	0.66	13626
weighted avg	0.92	0.78	0.82	13626

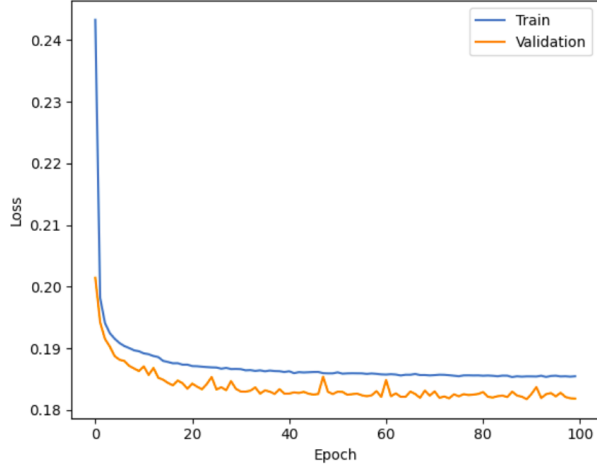
dataset. The initial architecture consisted of a fully connected feedforward network with one hidden layer, using **ReLU** (Rectified Linear Unit) activation functions for the hidden neurons and a single output neuron producing a probability through a **sigmoid** function. The network was optimized using the **Adam** optimizer, while the loss function adopted was **binary cross-entropy**. A preliminary experiment was conducted without any imbalance correction in order to observe the natural learning behavior of the network (Fig3). Although this configuration achieved a high overall accuracy (around 91%), it showed poor performance in identifying hazardous asteroids, confirming that the class imbalance problem significantly affects neural models as well (Tab4).

Table 4: Confusion matrix and Classification report for the initial Neural Network.

Confusion Matrix:
[[12244 56]
[1135 191]]

	precision	recall	f1-score	support
0	0.92	1.00	0.95	12300
1	0.77	0.14	0.24	1326
accuracy			0.91	13626
macro avg	0.84	0.57	0.60	13626
weighted avg	0.90	0.91	0.88	13626

Figure 3: Training and validation loss vs. epoch for the initial Neural Network.



To mitigate this limitation, class weighting was introduced during training. Class weights were automatically computed based on the distribution of the training data. This cost-sensitive learning strategy forced the network to focus more on the minority class and significantly improved the recall for hazardous asteroids, which approached values close to 0.99 in the test phase (Fig4, Tab5).

Figure 4: Training and validation loss vs. epoch for the Neural Network with class weighting.

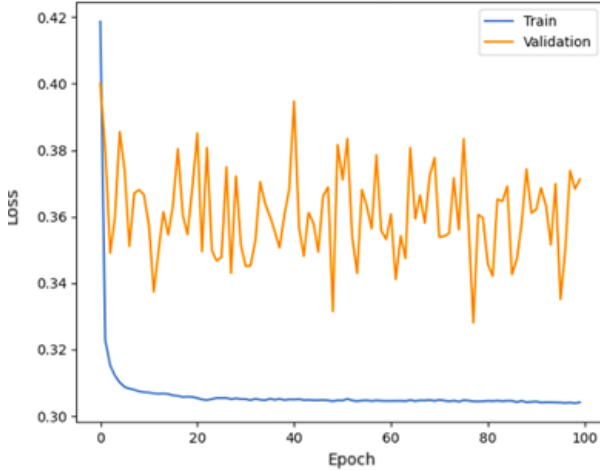


Table 5: Confusion matrix and Classification report for the Neural Network with class weighting.

Confusion Matrix:

```
[[9339 2961]
 [ 17 1309]]
```

	precision	recall	f1-score	support
0	1.00	0.76	0.86	12300
1	0.31	0.99	0.47	1326
accuracy			0.78	13626
macro avg	0.65	0.87	0.67	13626
weighted avg	0.93	0.78	0.82	13626

Several architectural configurations were subsequently explored, including networks with one and two hidden layers, in order to evaluate the impact of model

depth and neuron distribution on classification performance. Additionally, a Learning Rate Scheduler (**ReduceLROnPlateau**) was integrated to dynamically reduce the learning rate when the validation loss stopped improving. This mechanism stabilizes the training process and helps the model converge toward a more optimal solution (Fig5, Tab6).

Figure 5: Training and validation loss vs. epoch for the Neural Network with class weighting and Learning Rate Scheduler.

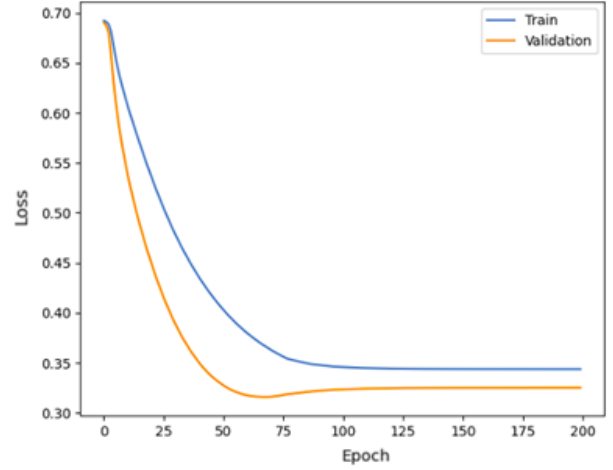


Table 6: Confusion matrix and Classification report for the Neural Network with class weighting and Learning Rate Scheduler.

Confusion Matrix:

```
[[9209 3091]
 [ 16 1310]]
```

	precision	recall	f1-score	support
0	1.00	0.75	0.86	12300
1	0.30	0.99	0.46	1326
accuracy			0.77	13626
macro avg	0.65	0.87	0.66	13626
weighted avg	0.93	0.77	0.82	13626

4.2 Results and Discussion

4.2.1 Results

The experimental results obtained from all tested models are summarized in the metrics table, which compares performance using several evaluation indicators, including Accuracy, F1-Score for the Hazardous class, Recall for Hazardous objects, Matthews Correlation Coefficient (MCC), and Cohen's Kappa. The comparison clearly highlights the strong impact of class imbalance on model behavior and shows how the introduction of balancing strategies significantly improves the detection of potentially dangerous asteroids. The custom implementation of Logistic Regression achieved a relatively high accuracy of 0.90, but the metrics associated with the minority class were extremely low (recall 0.01 and F1-score 0.03). This indicates that the model mostly predicts the

Table 7: Models Performance Summary.

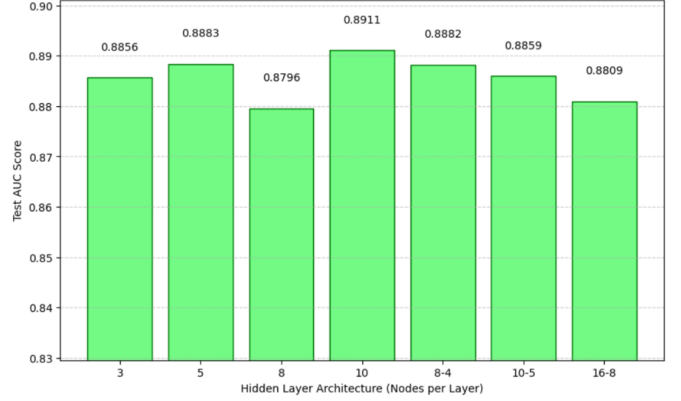
	Accuracy	F1-Score (Hazardous Class 1)	Recall (Hazardous Class 1)	MCC	Kappa
Logistic Regression (Custom)	0.90		0.03	0.01	0.05
Logistic Regression (Sklearn, Weighted)	0.78	0.46	0.93	0.45	0.36
NN (No Class Weighting)	0.91	0.24	0.14	0.31	0.22
NN (Weighted, 1 Hidden Layer)	0.78	0.47	0.99	0.48	0.38
NN (Weighted, 2 Hidden Layers)	0.79	0.48	0.98	0.48	0.38
NN (Weighted, LR Scheduler, 1 Hidden Layer)	0.77	0.46	0.99	0.47	0.36

majority class (“Safe”), producing high overall accuracy while failing to correctly identify hazardous asteroids. A substantial improvement appears when class weighting is applied in the Scikit-learn Logistic Regression implementation. In this configuration, the recall for hazardous objects increases dramatically to 0.93, while the F1-score rises to 0.46, reflecting a much better balance between precision and recall. As expected in imbalanced classification tasks, this improvement leads to a decrease in overall accuracy, which drops to 0.78. A similar pattern can be observed with the Neural Network trained without class weighting. Although this model reaches the highest overall accuracy (0.91), its ability to detect hazardous asteroids remains limited, with a recall of only 0.14. This result confirms that even more complex models tend to favor the dominant class when no corrective strategy is applied. Once balanced class weights are introduced, the Artificial Neural Networks show a significant improvement in identifying hazardous objects. The configurations with one and two hidden layers achieve extremely high recall values (0.99 and 0.98, respectively), while maintaining overall accuracy between 0.77 and 0.79. In addition, the MCC values (between 0.47 and 0.48) and Cohen’s Kappa scores (between 0.36 and 0.38) indicate a stronger predictive balance compared with the linear models (Tab7). Further insights are provided by the “AUC-ROC Score vs Neuron Configuration (Balanced Weights)” graph, which illustrates how the Neural Network architecture influences classification performance. The trend suggests that increasing the number of neurons in the hidden layers generally leads to higher **AUC-ROC** values, indicating a better separation capability between the two classes. However, after a certain level of architectural complexity, the improvements become marginal, suggesting a trade-off between model capacity and training stability (Fig6).

4.2.2 Discussion

RQ1: Can machine learning models effectively distinguish hazardous from non-hazardous asteroids using basic orbital and physical features? The confusion matrices and classification reports (Section 4.1, Figure 2, Table 5 and Table 6), along with the summary in Table 7, demonstrate that properly configured models can identify a large proportion of hazardous objects. This confirms the discriminative power of the selected features.

Figure 6: AUC-ROC Score vs Neuron Configuration.



RQ2: How does class imbalance affect model performance in asteroid hazard classification? As shown in the Logistic Regression baseline (Section 4.1.2, Table 2), the unbalanced model achieves high overall accuracy but fails almost completely in detecting hazardous asteroids. A similar pattern is observed with Neural Networks (Section 4.1.3, Table 4). Implementing class weighting, as defined in the Analytical Pipeline (Section 3.1), leads to a substantial improvement in recall, proving that class imbalance is the primary factor affecting model performance.

RQ3: Does increasing the depth of a Neural Network significantly improve the separation capability for hazardous objects compared to linear baseline models? The analysis of the AUC-ROC trends in Figure 6 shows that increasing the network depth provides some improvement in separation capability. However, these gains are relatively limited once the class imbalance is handled.

5 Conclusion

This study presented a machine learning framework for classifying Near-Earth Objects using physical and orbital parameters from the NASA NEO dataset. The experiments show that class imbalance represents the main challenge in asteroid hazard classification. Without corrective strategies, models tend to favor the majority class and fail to detect hazardous asteroids. By integrating class weighting techniques and focusing on recall optimization, both Logistic Regression and Artificial Neural Networks significantly improved their ability to identify potentially dangerous objects.

Future work may extend the dataset by incorporating temporal information such as historical close-approach records and orbital evolution data. Including these dynamic features could allow models to capture long-term trajectory patterns and further improve automated planetary defense monitoring systems.

References

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